THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which expression is a multiple of 5 for every whole number n? 1

(a) $5n + 1$ **(b)** $10n + 5$ **(c)** $10n + 2$ **(d)** $n + 5$

2. If $x = 5$ and $y = 2$, the value of $2x - 7y + xy$ is: 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

3. A rule is: output $= 3\times$ input -2. The output for the input 15 is: 1

(a) 42 **(b)** 45 **(c)** 43 **(d)** 44

4. If $x = 7$, the value of $4x^2 + 1$ is: 1

(a) 196 **(b)** 197 **(c)** 198 **(d)** 199

5. If $a = 6$ and $b = 3$, the value of $4a^2 - 8b$ is: 1

(a) 122 **(b)** 121 **(c)** 120 **(d)** 119

6. Simplify $9(x + 2) - 5(x - 3)$: 1

(a) $14x + 33$ **(b)** $4x + 3$ **(c)** $4x + 33$ **(d)** $4x - 3$

7. The perimeter of a rectangle with sides $8x + 2$ and $5x - 1$ is: 1

(a) $26x + 2$ **(b)** $6x + 6$ **(c)** $13x + 1$ **(d)** $80x$

8. If $n = 25$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 325 **(b)** 324 **(c)** 326 **(d)** 327

9. **Assertion (A):** The perimeter of a square of side s is 4s. 1
Reason (R): The perimeter is the sum of all four equal sides, s + s + s + s.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The expression 7n has the value 35 when n = 5. 1
Reason (R): A letter-number can take different values in different situations.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of $4x^2 - 2x + 6$ when $x = 3$. 2

12. Simplify: $6(x - 4) - 2(x - 7)$. 2

13. Write the sum of three consecutive multiples of 6, starting with $6n$, and simplify it. 2

14. Find the value of $5x^2 - 6x + 7$ when $x = 2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangle has length $(8x + 5)$ cm and breadth $(3x - 2)$ cm. Find its perimeter as an expression and its value for $x = 4$. 3

16. Write the sum of 7 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 3$. 3

17. Taxi A charges ₹10 plus ₹18 per km and taxi B charges ₹70 plus ₹8 per km. Write both fares for k km. For a journey of 8 km, which taxi is cheaper and by how much? 3

OR

A rectangle has length $(3x + 7)$ cm and breadth $(6x - 4)$ cm. Find its perimeter as an expression and its value for $x = 9$. 3

18. Write the sum of 8 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 2$. 3

OR

Taxi A charges ₹35 plus ₹16 per km and taxi B charges ₹95 plus ₹11 per km. Write both fares for k km. For a journey of 14 km, which taxi is cheaper and by how much? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan is r years old. His mother's age is 2 more than 3 times his age. His father is 6 years older than his mother. 5

(i) Write an expression for the mother's age.

(ii) Find the mother's age when $r = 12$.

(iii) Write an expression for the father's age.

(iv) Write and simplify an expression for the sum of the ages of Rohan, his mother and his father. Find its value when $r = 12$.

20. In a pattern, figure n has $8n + 5$ tiles. Find 5

(i) the tiles in figure 9,

(ii) the total tiles in figures 1 and 2, and

(iii) how many more tiles figure 9 has than figure 8.

OR

Priya has ₹x. Her brother has ₹25 more than twice her money. Write expressions for the brother's money and for their total money, and find both when $x = 60$. 5

END OF QUESTION PAPER — **Total: 40 Marks**